「Section 5.1 If $f: \mathbb{R} \to \mathbb{R}$ satisfies that

$$|f(x) - f(y)| \leq (x-y)^2 \quad \text{for all } x, y \in \mathbb{R}$$

Then f is constant.

Proof. Since $$\left| \frac{f(x) - f(y)}{x-y} \right| \leq |x-y| \quad \text{for all } x, y \in \mathbb{R},$$

given $x \in \mathbb{R}$, $f'(x)$ exists with $f'(x) = 0$.

Now, by the Mean value Theorem, for any $a < b$, there exists $c \in (a,b)$ s.t

$$f(a) - f(b) = (a-b) \cdot f'(c) = 0$$

Thus f is constant.

「Section 5.2. Let $f: (a,b) \to \mathbb{R}$ be given. If $f'(x) > 0$ for all $x \in (a,b)$, then f is injective.
Moreover, the inverse $f^{-1}: f[(a,b)] \to (a,b)$ exists and differentiable with

$$(f^{-1})'(f(x)) = \frac{1}{f'(x)} \quad (a < x < b)$$

Proof. Let $a < x < y < b$. Then, by the Mean-Value Theorem, there exists $c \in (x,y)$ s.t

$$f(y) - f(x) = f'(c) \cdot (y - x) > 0$$

Therefore, f is strictly increasing, thus injective. Now, the restriction

$$f: (a,b) \to f[(a,b)]$$

is bijection, thus there exists a inverse

$$f^{-1}: f[(a,b)] \to (a,b) : f(x) \mapsto x$$

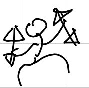
Now, $(f^{-1} \circ f)(x) = x$ implies that

「Section 5.3. Let $g: \mathbb{R} \to \mathbb{R}$ be given with $|g'| \le M$ for some $M > 0$.
Fix $\varepsilon > 0$, and define $f(x) = x + \varepsilon \cdot g(x)$. prove that f is injective if ε is small enough.

Proof. If $\varepsilon < \frac{1}{M}$, then $f'(x) = 1 + \varepsilon \cdot g'(x) > 0$. Therefore f is injective.

「Section 5.4. If $c_0, \cdots, c_n \in \mathbb{R}$ satisfies

$$c_0 + \frac{c_1}{2} + \frac{c_2}{3} + \cdots + \frac{c_{n-1}}{n} + \frac{c_n}{n+1} = 0$$



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, then the polynomial

$$c_0 + c_1 x + c_2 x^2 + \cdots + c_{n-1} x^{n-1} + c_n x^n = p(x)$$

has a real root in $(0,1)$.

Proof. Define $$q(x) = c_0 x + \frac{c_1}{2} x^2 + \cdots + \frac{c_{n-1}}{n} x^n + \frac{c_n}{n+1} x^{n+1}$$

Then, $q(0) = 0$ clearly, and $q(1) = 0$ by the assumption.
By the Mean-Value Theorem, there exists $\alpha \in (0,1)$ s.t

$$0 = q(1) - q(0) = q'(\alpha) \cdot (1-0) = q'(\alpha)$$

Directly, $p(x) = q'(x)$. therefore $q'(\alpha) = p(\alpha) = 0$.

「Section 5.5 Let $f: (0, \infty) \to \mathbb{R}$ be differentiable.
If $f'(x) \to 0$ as $x \to \infty$, then $f(x+1) - f(x) \to 0$ $x \to \infty$.

Proof. By the MVT, for any $x > 0$, there exists $c_x \in (x, x+1)$ s.t

$$f(x+1) - f(x) = f'(c_x) \cdot (1-0) = f'(c_x).$$

Let $\varepsilon > 0$ be given. Then, there exists $\delta > 0$ s.t $x \ge \delta \Rightarrow |f'(x)| < \varepsilon$.
For this $\delta > 0$,

$$|f(x+1) - f(x)| = |f'(c_x)| < \varepsilon$$

Thus proved.